

WEEKLY PROGRESS REPORT

Name	Arsh Kamal
Domain	Full Stack Development
Week	02
Quiz Score	90%

1. Overview

During Week 2 of the internship, I focused on learning core Java concepts and Object-Oriented Programming fundamentals. The learning material covered Java introduction and history, features, JDK and JRE, classes and objects, data types, variables, methods, operators, control-flow statements, loops, constructors and the this keyword. The second material covered inheritance, super and final keywords, abstraction, encapsulation, polymorphism, abstract classes and interfaces.

2. Achievements

- Studied the introduction, history and major features of Java.
- Understood the purpose of JDK and JRE.
- Learned core OOP concepts including class, object, inheritance, abstraction, encapsulation and polymorphism.
- Studied Java data types, variables, methods and operators.
- Learned decision-making, looping and branching control-flow statements.
- Studied constructors and the this keyword.
- Learned inheritance and its different types, along with the uses of super and final keywords.
- Studied abstraction through abstract classes and interfaces.
- Understood encapsulation using private data members and public setter/getter methods.
- Studied compile-time and runtime polymorphism, including method overloading and overriding.
- Completed the Week 2 quiz with a score of 90%.

3. Challenges and Hurdles

The main challenge during this week was understanding the differences and relationships between the OOP concepts, especially inheritance, abstraction, encapsulation and polymorphism. Differentiating method overloading from method overriding and understanding abstract classes and interfaces required careful revision. Reviewing the concepts helped improve my understanding.

4. Learning Resources

The two Java presentation materials provided as part of the internship were used as the primary learning resources. They covered Java fundamentals, OOP concepts and topics including inheritance, abstraction, encapsulation, polymorphism, abstract classes and interfaces.

5. Lessons Learned

- A class acts as a blueprint for an object, while an object has state and behaviour.

- Inheritance provides the concept of reusability.
- Abstraction hides implementation details while providing functionality to the end user.
- Encapsulation binds data members and member functions into a single unit and can restrict direct access to private data.
- Polymorphism allows a single action to be performed in different ways.
- Method overloading supports compile-time polymorphism, while overriding involves redefining a parent-class method in a child class.
- An abstract class cannot be instantiated, and an interface cannot be instantiated directly.

6. Next Week's Goals

- Continue revising the Java and OOP concepts learned during Week 2.
- Strengthen understanding of inheritance, abstraction, encapsulation and polymorphism.
- Practice applying the learned concepts in programming.
- Continue with the next internship learning activities and project-related requirements.

7. Additional Comments

Week 2 helped me build a stronger foundation in Java and Object-Oriented Programming. The 90% quiz score reflects my understanding of the topics covered during this week. I will continue to revise these concepts and apply my learning in the upcoming internship activities.